



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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**Let's Start!**

Is it easy to program a program?

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CRUD

Learning Outcome:

Uses the four basic functions in persistent storage to create the system. Talks about the four basic functions for persistent storage.

Can Do Statements:

- I can explain what the acronym CRUD means.
- I can talk about CRUD functions.
- I can describe the importance of CRUD for the user interface.

Ask your partner:

- Is there any basic code that all programs share? Which one?
- Is it easier to code a database in software or online? Why?





Let's Explore!

Paula is helping Ronald to find an error. What was it?

010create1011010
1010read0010110
0011update11001
1110delete001001

Instructions: Read the text below.

- Ronald:** Hi, Paula. I was wondering if you could actually help me with a little issue I have with this database I'm generating.
- Paula:** Sure, what's the deal?
- Ronald:** Well, I need to finish programming this database by tomorrow's class with Mr. Fredricksen, but I've been really busy with the other assignments and I really need some help.
- Paula:** Hmm, I suppose you want me to program that for you, don't you?
- Ronald:** Not really, I'd rather you help me with a fresh pair of eyes. As you can see, I've tried to make this program run during the whole day, but there's no use, it used to work at the beginning but now it's a total drag.
- Paula:** Let me take a glance over the codes... Ok, now I see, do you remember what CRUD stands for?
- Ronald:** Sure, Create, Retrieve, Update and Delete. Why do you ask?
- Paula:** Do you remember what they are for?
- Ronald:** Sure, they are by far the most important things to keep in mind when programming databases, as they permit programs to work well by allowing users to add, modify, import and delete data, as for example, a cell in a spreadsheet, but I wouldn't worry about that, that's something I master.
- Paula:** But, that's something I'd rather you checked twice. Would you mind looking at the codes and telling me what's missing?
- Ronald:** Oh my... It seems to be I forgot to add the R line while editing the codes. I've worked with these codes so many times that I completely forgot to check that; it's probably because I used to work with an old template that has that error.
- Paula:** Well, I suppose you've done it wrong all this time.
- Ronald:** I think I'd rather check the other projects.



**Let's Practice!**

Let's try understanding the meaning of CRUD!

Instructions: Underline all the sentences in past continuous.

1. CRUD have worked with four functions, they are: create, retrieve, update and delete.

2. He has never forget to use CRUD's function of Deleting when programming databases.

3. The function of Create have allowed me to add information in the spreadsheet.

4. The programmer have use CRUD to work with updated tools.

5. She have edited the codes by using CRUD to adapt them into the new template.

**Let's Practice!**

Let's check the CRUDE functions that Karl would use in his repository!

Instructions: Read the question. Select all the correct answers.

1. Which of the following CRUD functions would Karl use in his data repository?
 - a. Handle records and basic operations.
 - b. Data objects and handle records.
 - c. Memory Records.
 - d. Database tables.
 - e. Data objects.





Extension Activity

Let's write about CRUD functions!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Research about CRUD.
- Write an article about CRUD and its functions.
- Explain in your own words what the importance of CRUD for the user's interface is.
- Write five questions about CRUD and share them with your classmates.

Self-Assessment

Yes**No**

- I read about CRUD.
- I recognize CRUD's functions.
- I recognize the importance of CRUD for the user's interface
- I share CRUD information with my classmates.

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☐	☐
☐	☐
☐	☐



Target Words

- **glance** (noun): a quick view of something
e.g. Let me take a glance over the codes...
- **by far** (idiom): a huge or great degree of something
e.g. Sure, they are by far the most important things to keep in mind when programming databases.
- **drag** (noun): an annoying and obnoxious situation
e.g. It used to work at the beginning but now it's a total drag.
- **line** (noun): a row of words that edit a code
e.g. It seems to be I forgot to add the R line while editing the codes.
- **template** (noun): a standard format given to a text
e.g. I used to work with an old template that has that error.



Learning Points

Present perfect

We use **present perfect** to talk about things that happened in the past and ended in the present.

You can use present perfect to express ideas in any of the three following ways:

1. Ideas where time is not important

Sometimes you use **present perfect** to talk about the ideas in which you express lack of time, lack of importance of time for something, or ideas about something unfinished.

As time is not important for the action, you do not mention it in any of the ideas.

Example:

- *Code lines have been important to edit software.*
- *The IT personnel have fixed the software.*

INSERTAR IMAGE 1. (Image1)

2. Repeated ideas

Every time actions are done repeatedly in the past but are somehow related to the present, you express them with **present perfect**.

Example:

- *I've edited the code lines several times!*
- *We've added different code lines every time we try to fix the problem.*

INSERTAR IMAGE 2. (Image2)

3. Long unfinished actions.

If you started an action in the past and is still in progress, you can use **present perfect** to express such idea.

Example:

- *The IT has struggled with that code since he got in charge of modifying the program.*

INSERTAR IMAGE 3. (Image3)

In order to make a sentence in present perfect you have to use the following structure:

Subject + have/has + past participle verb + complement.

Note: you can contract the Auxiliary verb have/has as follows





Learning Points

Subject	Aux	+ Contraction
I	Have	I've
You		You've
He	Has	He's
She		She's
It		It's
We	Have	We've
They		They've

Examples

- *I've been really busy with the other assignments and I really need some help.*
- *I've tried to make this program run during the whole day.*
- *I've worked with these codes so many times that I completely forgot to check that.*

CRUD acronym

CRUD stands for **Create, Retrieve, Update** and **Delete**

When you are working with databases, it is common to talk about **CRUD** procedures as it is said that if the program does not have the whole four components (**create, retrieve, update** and **delete**) the software is not complete; therefore, it won't work.

Each procedure has to be edited and created using code lines. **CRUD** procedures can be created in specialized software and apps that can be either free or privative such as MySQL or Oracle among many others.

Each of the letters of **CRUD** represents one specific procedure:

The **Create** procedures allow the user to **insert data** into a software that can be either installed in a PC or online regarding to any topic. It can be either a name, an address or email, among many others.

The **Read** procedures allow the user to select or read the cells containing the information previously added.

The **Update** procedures allow the users to modify the data previously added in the **C** procedures.

The **Delete** procedures allow the users to delete rows of information any time the user considers this necessary.

